

Assignment 2: Mathematical Properties and Computation of SVD

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1 Theoretical Proofs regarding $S = A^T A$

Let $A \in \mathbb{R}^{m \times n}$ denote an arbitrary linear mapping matrix. We analyze the algebraic foundations of the symmetric operator defined as $S = A^T A$.

1.1 Symmetry Verification ($S = S^T$)

To verify that the operator S is inherently symmetric, we evaluate its transposed state leveraging the distributive properties of matrix transposition, where $(BC)^T = C^T B^T$:

$$S^T = (A^T A)^T = A^T (A^T)^T$$

Since the double transposition operation returns the matrix to its primary state, $(A^T)^T = A$, we substitute directly:

$$S^T = A^T A = S$$

This completes the formal verification showing that S equals its own transpose.

1.2 Positive Semi-Definite Property

An operator S is classified as positive semi-definite if the inner product form yields a non-negative scalar for any arbitrary non-zero vector $x \in \mathbb{R}^n$. Expanding the definition:

$$x^T S x = x^T (A^T A) x = (x^T A^T)(Ax)$$

Using the identity $(Ax)^T = x^T A^T$, we rewrite the expression as an inner product of a single transformed vector:

$$x^T S x = (Ax)^T (Ax) = \|Ax\|_2^2$$

Because the squared Euclidean norm of any vector in a real vector space is fundamentally bounded below by zero ($\|v\|_2^2 \geq 0$), it strictly follows that:

$$x^T S x \geq 0 \quad \forall x \in \mathbb{R}^n$$

Thus, the matrix S is proven to be positive semi-definite, ensuring that its eigenvalues are real and non-negative.

2 Numerical Experimentation

Using the NumPy linear algebra pipeline, the singular value decomposition was executed on a custom 3×2 matrix. The empirical outputs validated that the singular values σ_i match the square roots of the computed eigenvalues from the internal symmetric system S .